MATH329 · OPERATIONS RESEARCH

Midterm 462 Solutions

09/03/2025 · 30 marks

Muharib

01 Solution Q1: theory

a The special cases of the graphical method

1. Alternative optimal solutions
2. Unbounded solution
3. No feasible solution

b The two requirements to start simplex

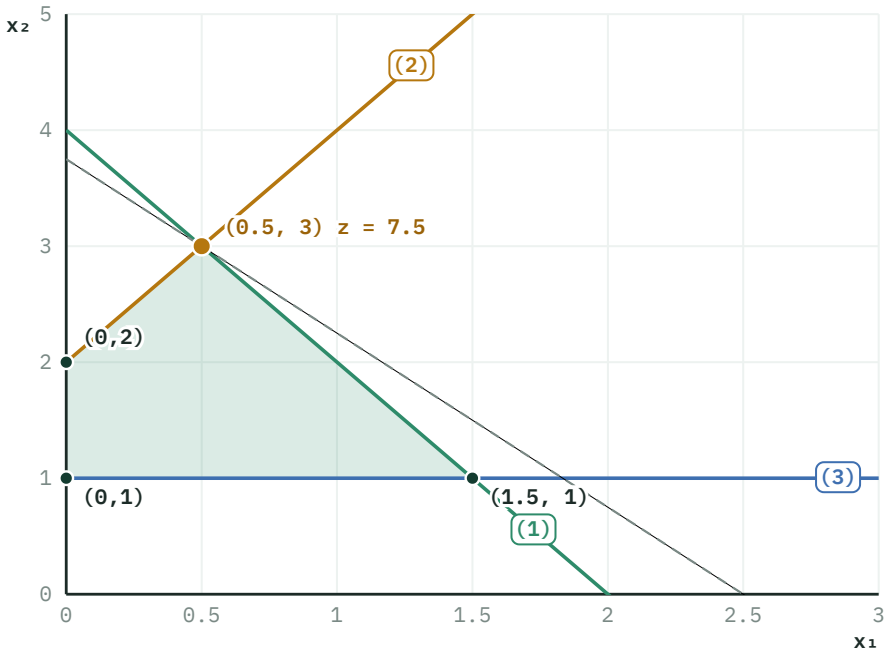
1. The coefficients of the basic variables (slack variables) form an **Identity matrix**.
2. In the objective function row, the coefficients of the basic variables are **zero**.

02 Solution Q2: graphical

1 Prepare the constraints

(1)	$2x_1 + x_2 = 4$	points (0, 4) and (2, 0)
(2)	$-2x_1 + x_2 = 2$	points (0, 2) and (−1, 0)
(3)	$x_2 = 1$	horizontal line

Constraints (1) and (2) hold at $(0, 0)$, so their region is on the origin's side.
Constraint (3) is $\geq$, so its region is above the line $x_2 = 1$.



2 The corner points

Corner	Where it comes from	$z = 3x_1 + 2x_2$
(0, 1)	(3) with the x_2 axis	2
(1.5, 1)	(1) with (3)	6.5
(0.5, 3)	(1) with (2)	**7.5**
(0, 2)	(2) with the x_2 axis	4

(1) + (2): $2x_2 = 6 \rightarrow x_2 = 3$ $\quad$ $2x_1 + 3 = 4 \rightarrow x_1 = 0.5$

Z = 7.5 at $x_1 = 0.5$ and $x_2 = 3$

03 Solution Q3: standard form

1 The methods 2 M

Graphical method and Algebraic method.

2 Standard form (6 M)

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

$x_2 = x_2^+ - x_2^-$ in every place x_2 appears

Maximum $Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$	
S.to	$2x_1 + x_2^+ - x_2^- + s_1 = 12$
	$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$
	$-4x_2^+ + 4x_2^- = 7$
	$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

Marking: two marks for the objective function and one for each of the four lines.

04 Solution Q4: simplex for a min problem

a Standard form (2 M)

Minimize $Z = x_1 - 2x_2 + 0s_1 + 0s_2$	
S.to	$2x_1 + 3x_3 + s_1 = 1$
	$3x_1 + 2x_2 - x_3 + s_2 = 5$
	$x_1, x_2, x_3, s_1, s_2 \geq 0$

The key writes the last line as $x_1, x_2, s_1, s_2, s_3 \geq 0$, which is a typo: the problem has x_3 and no s_3.

b Simplex (8 M)

The Z row comes from $z - x_1 + 2x_2 = 0$. The problem is min, so the entering variable is the most positive entry.

Basic	x_1	x_2	x_3	s_1	s_2	Sol.	Ratio
Z	-1	2	0	0	0	0	
s_1	2	0	3	1	0	1	–
s_2	3	**2**	-1	0	1	5	$\frac{5}{2}$
Z	-4	0	1	0	-1	-5	
s_1	2	0	**3**	1	0	1	$\frac{1}{3}$
x_2	$\frac{3}{2}$	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{5}{2}$	–
Z	$-\frac{14}{3}$	0	0	$-\frac{1}{3}$	-1	$-\frac{16}{3}$	
x_3	$\frac{2}{3}$	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	
x_2	$\frac{11}{6}$	1	0	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{8}{3}$	

Iteration 1: the most positive entry is 2 under x_2. The s_1 row has a zero coefficient, so it has no ratio, and the s_2 row has ratio $\frac{5}{2}$. So s_2 leaves and the pivot is 2.

Iteration 2: a positive 1 remains under x_3. The s_1 row has ratio $\frac{1}{3}$, and the x_2 row has a negative coefficient. So s_1 leaves and the pivot is 3.

Optimum $z = -\frac{16}{3} = -5.33$ at $x_1 = 0$, $x_2 = \frac{8}{3} = 2.67$, $x_3 = \frac{1}{3} = 0.33$

Check: $z = 0 - 2(\frac{8}{3}) = -\frac{16}{3}$

The same tableaus in decimals (as in the answer key)

Basic	x_1	x_2	x_3	s_1	s_2	Sol.
Z	−1	2	0	0	0	0
s_1	2	0	3	1	0	1
s_2	3	**2**	−1	0	1	5
Z	−4	0	1	0	−1	−5
s_1	2	0	**3**	1	0	1
x_2	1.5	1	−0.5	0	0.5	2.5
Z	−4.67	0	0	−0.33	−1	−5.33
x_3	0.67	0	1	0.33	0	0.33
x_2	1.83	1	0	0.17	0.5	2.67

First table 2m (every row 0.5m + 0.5m for pivot element).

Second and third table 3m (1m for every row). Every mistake in calculation −0.5m.